Model Reduction for Nonlinear DABNet Models

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Abstract. In this paper we present an identification and reduction technique suitable for a particular nonlinear model structure. We use the DABNet structure, which is composed of a linear dynamic system followed by a nonlinear static map. The linear dynamic system is initially spanned by a set of discrete Laguerre systems, and then cascaded with a single hidden layer Perceptron. A linear model reduction technique is performed on the hidden nodes of the neural network as part of the identification process. In that way, it is possible not only to identify the main time constants, but also to reduce the dimensionality of the Perceptron input. Results concerning the application of the methodology to the approximation of a polymer process are presented.

Keywords: model reduction, nonlinear modeling, non-linear identification, state space models

1. Introduction

This paper presents a practical approach for identifying low order nonlinear dynamic models from process data, particularly oriented to implementing a Model Predictive Control (MPC) scheme. In the identification of a general nonlinear model it is essential to deal with the issues of ensuring long-term prediction capabilities, a feasible and practical identification procedure and, model suitability for certain kinds of processes. We address these problems in order to make the nonlinear empirical model more reliable.

The first important issue to be considered is ensuring the long-term prediction capabilities of the model. This is evident in implementations of MPC techniques, one of the most successful process control approaches (Morari and Lee, 1997). In MPC, a dynamic model is used to predict the behavior of the system over a future time horizon in response to a sequence of input changes. Stability is achieved by choosing the horizon large enough, i.e., larger than the maximum time required to steer the system to the origin from any state (Michalska and Mayne, 1993). In many systems this horizon may be large, so models should be able to predict over large horizons without degrading the prediction performance.

A second relevant point is getting a feasible and practical identification procedure. Most industrial processes that apply MPC employ linear step-response models or finite impulse response models (FIR). Although these approaches are widely and successfully used in industry, they require extensive plant testing which often results in over parametrization for the models (Qin and Badgwell, 1996). For **example** the dynamics of a first order SISO process, which can be represented with only 3 parameters (gain time constant, and dead-time), typically requires 30 to 120 coefficients in a non-parametric model. This is even worst in the nonlinear case, i.e. the Volterra Series. Alternative MPC implementations using Recurrent Neural Networks (Chen and Weingand, 1994) provide a reasonable solution to long-term prediction problems, but training such models is very difficult. So, efficient and feasible training methods are needed.

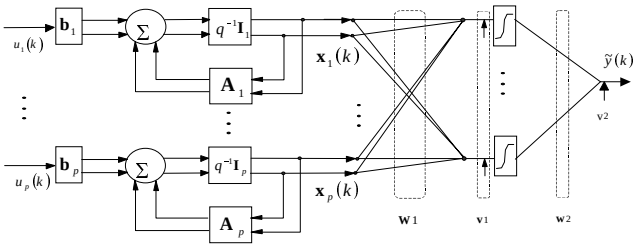


Figure 1 *DABNet* Structure

In this paper, we present a methodology to obtain models for a class of nonlinear systems in terms of a linear dynamic part followed by a nonlinear static map using input-output data. The linear dynamic system is initially spanned by a set of discrete Laguerre systems, and then cascaded with a single hidden layer Perceptron as its depicted in Figure **2**. In previous work (Sentoni *et al.*, 1998), it was shown that a nonlinear combination of discrete Laguerre systems is able to approximate every multi-input nonlinear discrete system having fading memory. After the initial span of the linear layer by Laguerre systems, a model reduction technique (a linear balancing, Zhou *et al.*, 1996) is performed on the hidden nodes of the neural network as part of the identification process. The balancing is performed in such a way that the linear state space representation is de-coupled by blocks. Doing so, it is possible not only to identify the main time constants, but also to reduce the dimensionality of the Perceptron input space. In addition, the non-controllable/ non-observable modes are removed from the model. The final *DABNet* model (De-coupled A-B matrices Neural Network) consists of a sparse linear state space system whose states, de-coupled by blocks, are mapped by a neural network. This de-coupled representation provides insight into the final model, especially since, in a practical

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identification scheme, the individual state space matrices can themselves be represented as loosely coupled first and second order sections. The *DABNet* structure is well suited to be embedded into a MPC Strategy, providing better long-term predictions than a NARX model in which the output errors are fed back into model input and amplified (Tsoi and Back, 1994). In addition, *DABNet* models show all of the following features: long-term prediction capabilities with minimal parameterization, practical identification that derive information for process control and ability to identify a large class of real world processes.

This paper is structured as follows. In Section 2 the nonlinear model representation is presented, while in Section 3 is presented the bulk of the identification and order reduction process and a summary of the methodology. In Section 4 the proposed methodology is applied to the approximation of a continuous polymerization process, finishing with some conclusions and the future work arising from this study.

2. Nonlinear Model Representation

We are interested in approximating the class of nonlinear, causal, discrete time-invariant, multiple-input single-output fading memory systems using the structure shown in Figure **1.** An operator that maps input signals into output signals can represent a system. Additionally, if we are interested in discrete systems, those signals become discrete sequences. To clarify the notation, matrices are denoted in bold uppercase while functions are denoted in italic lowercase if they are scalars or in bold its vector valued. Thus, **u** represents a vector or a vector sequence in accordance with the context, while $\mathbf{u}(k)$ represents a value of the sequence at the particular time *k*. The structure showed in Figure **1** is initially spanned using Laguerre systems as depicted in Figure **2**.

2.1 Laguerre Systems

The advantage of the Laguerre systems is that they provide a stable parameterization for a linear system by using orthonormal basis functions generated with the same pole. For a system with *p* inputs a set of Laguerre systems can defined in the complex z-plane as:

$$L_i^j(\cdot) = \frac{\sqrt{\eta_j}\, z^{-d_j+1}}{z-a_j}\left[\frac{1-a_j z}{z-a_j}\right]^i,\ i:0,\ldots,\infty,\ j:1:,..,p, \quad (1)$$

where $L_i^j(z)$ is the *Z*-transform of $l_i^j(k)$, the *i*-th order discrete Laguerre system for the *j*-th input component; a_j ($|a_j|<1$) the generating pole associated to the *j*-th input; $\eta_j = 1-a_j^2$; $d_j = 1$, unless there is a pure time delay for the *j*-th input for which $d_j > 1$. It is well known that a linear combination of Laguerre systems can approximate a great variety of linear systems (Heuberger *et al*, 1993; Wahlberg, 1991; Gu *et al*, 1989). The main feature in those schemes is that the linear system can be expressed as a linear

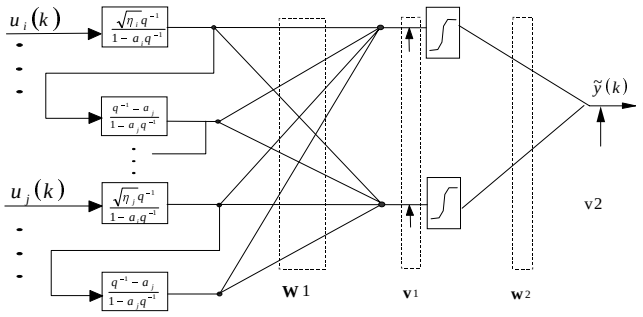


Figure 2: *DABNet* Initialization.

combination of orthonormal systems generated with the same pole. The only necessary previous knowledge of the systems is reflected in choosing this pole a_j, closely related to its main time constant τ_j (Zervos *et al*, 1988):

$$a_j = 1 - T/\tau_j \quad (2)$$

The dominant time constant τ_j relates the output with the *j*-th input while *T* is the sampling time. In this approach, augmenting the number of Laguerre systems may easily increase the identification structure. In addition, the identification task is robust respect to the choice of sampling time *T*, as well as to the choice of model order (Wahlberg, 1991). Notice, that the set (1) has an infinite number of terms, and for a practical application this number should be finite. To do this we take m_j Laguerre systems relating the output to the *j*-th input generated with the pole a_j, for $j:1:,..,p$. Next, we need to express this set in a state space form for then applying a model order reduction technique. Here we represent a linear dynamic system in a general way, by the quadruple of matrices $[\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}]$ defining its dynamics. Therefore, the quadruple $[\mathbf{A}_j,\mathbf{b}_j,\mathbf{C}_j,\mathbf{0}]$ denotes the m_j systems associated with the *j*-th input expressed in state space form. Those matrices can be expressed as the composition of a low-pass section, $\sqrt{\eta_j}(z-a_j)^{-1}$, in series with (m_j-1) all-pass sections $(1-za_j)/(z-a_j)$. Possible realizations are $[a_j, \sqrt[4]{a_j^2\eta_j}, \sqrt[4]{a_j^2\eta_j}, 0]$ for the low-pass and $[a_j, \sqrt{\eta_j}, \sqrt{\eta_j}, -a_j]$ for the all-pass systems respectively. Then, the state space representation $[\mathbf{A}_j,\mathbf{b}_j,\mathbf{C}_j,\mathbf{0}]$ is obtained by performing simple systems operations (see for example pp. 65 of Zhou et al, 1996). The outputs of the whole set of the Laguerre systems $\hat{\mathbf{y}}(k)$ are the outputs of the following system expressed in state space from:

$$\begin{aligned}\hat{\mathbf{x}}(k+1) &= \hat{\mathbf{A}}\hat{\mathbf{x}}(k) + \hat{\mathbf{B}}\mathbf{u}(k)\\ \hat{\mathbf{y}}(k) &= \hat{\mathbf{C}}\hat{\mathbf{x}}(k)\end{aligned}, \quad (3)$$

where $m = \sum_{j=1}^{p} m_j$, $\hat{\mathbf{A}}: m\times m$, $\hat{\mathbf{B}}: m\times p$ and $\hat{\mathbf{C}}: m\times m$. Notice the particular structure of the $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ matrices:

$$\hat{\mathbf{A}}=\begin{bmatrix}\mathbf{A}_1 & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \ddots & \cdots & \mathbf{0} & \mathbf{0}\\ \vdots & \vdots & \mathbf{A}_j & \vdots & \vdots\\ \mathbf{0} & \mathbf{0} & \cdots & \ddots & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{A}_p\end{bmatrix},\ \hat{\mathbf{B}}=\begin{bmatrix}\mathbf{b}_1 & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \ddots & \cdots & \mathbf{0} & \mathbf{0}\\ \vdots & \vdots & \mathbf{b}_j & \vdots & \vdots\\ \mathbf{0} & \mathbf{0} & \cdots & \ddots & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{b}_p\end{bmatrix} \quad (4)$$

showing a balanced realization with independent states $\hat{\mathbf{x}}_j(k):1\times m_j$ (de-coupled by blocks). In a linear system, these states are linearly combined to obtain a linear model. In the *DABNet* structure a single hidden layer Perceptron nonlinearly maps these states.

A single hidden layer Perceptron $NN(\cdot)$ characterized by $(\mathbf{w}2, v2, \mathbf{W}1, \mathbf{v}1)$, perhaps the most used neural network type, completes the *DABNet* structure by mapping the states of *Eq.* 3 into the output $\tilde{y}(k)=NN[\hat{\mathbf{x}}(k)]$:

$$\tilde{y}(k)=\mathbf{w}2\Phi\left[\mathbf{W}1\,\hat{\mathbf{C}}\,\hat{\mathbf{x}}(k)+\mathbf{v}1\right]+v2 \quad (5)$$

$\mathbf{W}1:[n1\times m]$ is the input weight matrix with $n1$ neurons; $\mathbf{v}1:[n1\times 1]$ is the input bias vector; $\mathbf{w}2:[1\times n1]$ is the output weight vector, and $v2$ is the output bias. The activation function $\Phi(s_1,s_2,...,s_{n1})=(\phi(s_1),\phi(s_2),\ldots ...,\phi(s_{n1}))$ is any permitted sigmoidal activation. Then, we have the following theorem.

Theorem: DABNet Structure

Let N be any time invariant operator with fading memory in a subset $\boldsymbol{K}$ of the input domain. Then, given any $\varepsilon>0$, there are a set of Laguerre operators $\left\{L_1^1(\cdot),\cdots,L_{m_1}^1(\cdot),\cdots,L_1^p(\cdot),\cdots,L_{m_p}^p(\cdot)\right\}$ and a single hidden layer Perceptron $NN(\cdot)$ characterized by $(\mathbf{w}2, v2, \mathbf{W}1, \mathbf{v}1)$ such that for all $\mathbf{u}\in\boldsymbol{K}$

$$\left\|N\mathbf{u}-\hat{N}\mathbf{u}\right\|<\varepsilon, \quad (6)$$

where $\hat{N}\,\mathbf{u}(k)=NN\left(L_1^1\mathbf{u}(k),\cdots,L_{m_1}^1\mathbf{u}(k),\cdots,L_1^p\mathbf{u}(k),\cdots,\cdots,L_{m_p}^p\mathbf{u}(k)\right)$ (Sentoni *et al.*, 1998).

This theorem ensures that a certain class of non-linear systems can be approximated by a "nonlinear combination" of Laguerre systems to any degree of accuracy $\varepsilon>0$. This approximation holds over a bounded input signal domain $\boldsymbol{K}$, in good agreement with any physical system. A Single Hidden Layer Perceptron $NN(\cdot)$ characterized $(\mathbf{w}2, v2, \mathbf{W}1, \mathbf{v}1)$ performs the nonlinear combination. The use of this neural network allows the utilization of order reduction techniques for pruning the dimension on the linear part, initially spanned by the Laguerre systems. This completes the description of the nonlinear model.

3. Identification and Model Reduction

In this section, we present how to reduce the dimension of the linear dynamic part of the *DABNet* structure by keeping the useful de-coupled matrix structure. In a first step, the model is adjusted using Laguerre systems because of the little knowledge that is required from a dynamic point of view. This is important to avoid a complex nonlinear fitting algorithm, like traditional ones for these kind of structures (Boutayeb and Darouach, 1995). However, this is at a small-added cost: the dimension of the state vector $\hat{\mathbf{x}}(k)$ is bigger than if were using all the (unknown) dynamic information. What can be done to reduce the dimension of this vector? Let us rewrite the set of equations for the *DABNet* model:

$$\hat{\mathbf{x}}(k+1)=\hat{\mathbf{A}}\hat{\mathbf{x}}(k)+\hat{\mathbf{B}}\mathbf{u}(k) \quad (7.1)$$

$$\tilde{\mathbf{y}}(k)=\mathbf{w}2\Phi\left(\hat{\mathbf{W}}1\,\hat{\mathbf{x}}(k)+\mathbf{v}1\right)+v2 \quad (7.2)$$

where $\hat{\mathbf{W}}1=\mathbf{W}1\,\hat{\mathbf{C}}=\left[\hat{\mathbf{W}}1_1\ldots\hat{\mathbf{W}}1_j\ldots\hat{\mathbf{W}}1_p\right]$, $\hat{\mathbf{W}}1_j:n1\times m_j$ for $n1$ hidden neurons and m_j Laguerre systems. Notice that the system $\left[\hat{\mathbf{A}}_j,\hat{\mathbf{b}}_j,\hat{\mathbf{W}}1_j,\mathbf{0}\right]$ forms a linear system with one output for every hidden neuron, over which the sigmoidal activation function is applied. Because this system is linear, all the theory of linear model reduction can be applied. In that way, the final model is obtained after balancing and truncating the linear system $\left[\hat{\mathbf{A}}_j,\hat{\mathbf{b}}_j,\hat{\mathbf{W}}1_j,\mathbf{0}\right]$ to reduce the dimension of the system. The objective of this step is to reduce the dimension of the vector $\hat{\mathbf{x}}_j(k)$, retaining the highest possible quantity of information in the reduced system. One way to do this is to generate a balanced realization of the linear system $\left[\hat{\mathbf{A}}_j,\hat{\mathbf{b}}_j,\hat{\mathbf{W}}1_j,\mathbf{0}\right]$ (Zhou *et al*, 1996) and then truncate it.

The observability Grammian $\mathbf{P}_j$ and the controllability Grammian $\mathbf{Q}_j$ (for $j:1,...,p$) are matrices that solve the following discrete-time Lyapunov equations:

$$\mathbf{P}_j-\left(\hat{\mathbf{A}}_j\right)^T\mathbf{P}_j\,\hat{\mathbf{A}}_j=\left(\hat{\mathbf{W}}1_j\right)^T\hat{\mathbf{W}}1_j \quad (8)$$

$$\mathbf{Q}_j-\hat{\mathbf{A}}_j\,\mathbf{Q}_j\,\left(\hat{\mathbf{A}}_j\right)^T=\hat{\mathbf{b}}_j\,\left(\hat{\mathbf{b}}_j\right)^T \quad (9)$$

uniquely for the linear system $\left[\hat{\mathbf{A}}_j,\hat{\mathbf{b}}_j,\hat{\mathbf{W}}1_j,\mathbf{0}\right]$ (Zhou *et al*, 1996). Additionally, there exists a similarity transformation $\mathbf{T}_j$ (Zhou *et al*, 1996) in which the system is in balanced form, that is:

$$\mathbf{P}_{j_{bal}}=\mathbf{Q}_{j_{bal}}=\Sigma_j=diag\left(\sigma_{j1},...,\sigma_{jm_j}\right); \quad (10)$$

where $\sigma_1\geq...\geq\sigma_m$ are the Hankel singular values. Now, a balanced realization can be computed simply as

$$\mathbf{A}_{j_{bal}}=\mathbf{T}_j\hat{\mathbf{A}}_j\left(\mathbf{T}_j\right)^T,\ \hat{\mathbf{b}}_{j_{bal}}=\mathbf{T}_j\,\hat{\mathbf{b}}_j,\ \hat{\mathbf{W}}1_{j_{bal}}=\hat{\mathbf{W}}1_j\left(\mathbf{T}_j\right)^T \quad (11)$$

Small Hankel values ($\sigma_{jr}>>\sigma_{jr+1}$) indicate that the system can be reduced to an r- order model by direct truncation as:

$$\overline{\mathbf{A}}_j=\left[\mathbf{I}_r\,0\right]\mathbf{A}_{j_{bal}},\ \overline{\mathbf{b}}_j=\left[\mathbf{I}_r\,0\right]\overline{\mathbf{b}}_{j_{bal}},\ \overline{\mathbf{W}}1_j=\mathbf{W}1_{j_{bal}}\begin{bmatrix}\mathbf{I}_r\\0\end{bmatrix} \quad (12)$$

The final reduced system is:

$$\begin{aligned} \bar{\mathbf{x}}(k+1) &= \bar{\mathbf{A}}\,\bar{\mathbf{x}}(k) + \bar{\mathbf{B}}\,\mathbf{u}(k) \\ \bar{\mathbf{y}}(k) &= \bar{\mathbf{w}}2\;\Phi\left(\bar{\mathbf{W}}1\;\bar{\mathbf{x}}(k) + \bar{\mathbf{v}}1\right) + \bar{v}2 \end{aligned} \tag{13}$$

where $\bar{\mathbf{W}}1 = \left[\bar{\mathbf{W}}1_1 \ldots \bar{\mathbf{W}}1_j \ldots \bar{\mathbf{W}}1_p\right]$. The matrices $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, have the same structure than $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ in *Eq.* 4**,** are built up using $\bar{\mathbf{b}}_j$, $\bar{\mathbf{A}}_j$ from *Eq.* 12.

A more reduced model can be obtained if the balancing and reduction procedure is applied to $\left[\hat{\mathbf{A}}, \hat{\mathbf{B}}, \hat{\mathbf{W}}1, \mathbf{0}\right]$, that is considering the global dynamics. However, doing so, the structure in the matrices $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ is lost: these matrices become full matrices.

3.1 Proposed Strategy: Summary

1. Expand the System into a Laguerre Basis
   - Select m_j Laguerre systems (Eq. 1) for each input, typically $5 \le m_j \le 10$ are enough.
   - Select each pole of the Laguerre basis to be determined as a_j (*Eq.*).
   - Calculate the state space representation $\left[\hat{\mathbf{A}}, \hat{\mathbf{B}}, \hat{\mathbf{C}}, \mathbf{0}\right]$ for the whole set of Laguerre systems (*Eq.* 3).
2. Train the Neural Network
   - Given *N* samples $\left(\mathbf{u}(k), y(k)\right)$ drawn from the system, evaluate the Laguerre states $\tilde{\mathbf{x}}(k)$ (*Eq.* 7.1).
   - With $\tilde{\mathbf{x}}(k)$ as input and the output $y(k)$, fit the model to get the parameters $\left(\mathbf{w}2, \mathbf{b}2, \hat{\mathbf{W}}1, \mathbf{b}1\right)$ (*Eq.*7.2).
3. Balance the Model over the Hidden Nodes
   - Find $\mathbf{P}_j$ and $\mathbf{Q}_j$ Grammians by solving the Lyapunov Equations (8) and (9).
   - Find $\mathbf{T}_j$ that makes the system balanced $\mathbf{P}_{j\,bal} = \mathbf{Q}_{j\,bal}$, and the singular values $\left(\sigma_{j_1}, \ldots, \sigma_{j_{m_j}}\right)$ (*Eq.*10).
   - Find $\mathbf{A}_{j\,bal}$, $\hat{\mathbf{b}}_{j\,bal}$, $\hat{\mathbf{W}}1_{j\,bal}$ that balance the system (*Eq.* 11).
4. Truncate the model
   - If $\sigma_{j_r} >> \sigma_{j_{r+1}}$ (typically $\sigma_{j_r} > 10\,\sigma_{j_{r+1}}$), reduced to an *r*- order model by direct truncation using *Eq.* 12.
5. With the new dynamics $\left(\bar{\mathbf{A}}, \bar{\mathbf{B}}\right)$, retrain the neural network for the best fit to get the model of Eq. 13.

4. Application Example

In this section, we illustrate the feasibility of the identification and model reduction methodology applied to identifying the conversion factor x_p (*Eq.* 14) of a continuous polymerization reactor. The process consists of a polymerization solution of methyl-methacrylate (MMA), initiated by azo-bis-isobutotyronitrile (AIBN), in which toluene is used as the solvent. Species balances for the monomer, initiator, solvent and dead polymer (assuming perfect mixing) and an energy balance for the reactor give the set of ordinary differential equations (Soroush and Kravaris, 1992):

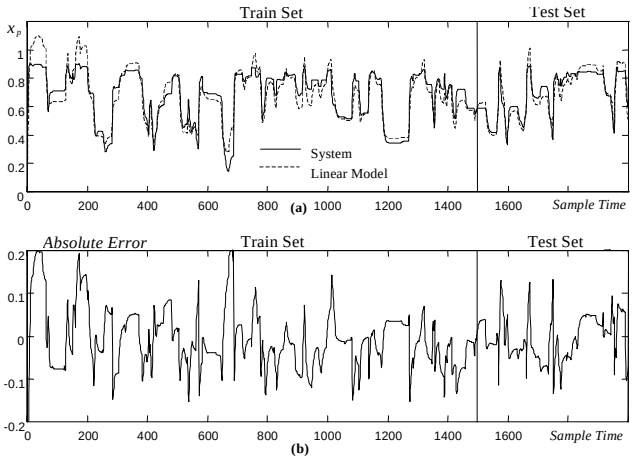


Figure 3. **a)** Linear Model of the Conversion Factor
b) Approximation Error of the Linear Model

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{f}\left(\mathbf{x}(t), \mathbf{u}(t)\right) \\ x_p &= h\left(\mathbf{x}(t)\right) \end{aligned} \tag{14}$$

The state variable $\mathbf{x} = \left[C_m, C_i, C_s, \mu_1, T, T_j\right]^T$ accounts for: C_m, concentration of the monomer; C_i, concentration of the initiator; C_s, concentration of the solvent; μ_1: mass concentration of the polymer chains; *T*, reactor temperature; and T_j jacket temperature. The input vector $\mathbf{u}(t) = \left[P, F_{cw}, F_i\right]^T$ is composed by P, the heater input power; F_{cw}, the cooling water inlet flow rate; and F_i, the initiator flow rate stream.

The aim of this example is to identify a DABNet model of the conversion factor of the polymer using the power given to the heater, the cooler flow rate and the initiator flow rate as inputs. Some data (input/output sequences) has to be generated to identify the system. The input sequence utilized to perform the identification is generated using the approach of Mane and Doyle (1997). The input sequence is allowed to change only at the end of a sampling time. If the input is changed, its new values are drawn from a uniform distribution with a probability P_s, where the switching factor P_s corresponds to the probability that the input changes at the end of any sampling interval. This probability P_s is adjusted to provide a signal that excites the system evenly over the frequency range of interest. Once the input sequences are available, they are used to get the output sequence by integrating the Equations (*Eq.* 14). In this example, two data sets were generated using a probability P_s =0.04 to identify the system: a training of 1500 samples and test set of 500 samples. The output sequences were generated integrating those inputs using the set of equations (14) with a sampling time of T = 1000 seconds. The training set was used to fit the model, while the test set was used to cross-validate the model. Using the training set, a DABNet model and a linear model were

identified.

A six states linear state space model was adjusted using a partial least square strategy for comparison proposes. It can be appreciated from Figure **3**, that a linear model is not good enough to approximate the system over the whole range. The identification process for the DABNet models begins choosing the poles of the of the Laguerre systems. As a way to show the relative robustness of the approach against choosing bad values for the Laguerre poles, those were selected to be $a_1 = a_2 = a_3 = 0.5$ (just in the middle of the valid range), while the dead-times were selected to be: $d_1 = 1, d_2 = 1, d_3 = 1$. For the initial expansion, six Laguerre basis terms were used for each input, resulting in a matrix $\hat{\mathbf{A}}$ (Eq. 4) of $[18 \times 18]$. For the neural network, we arbitrarily choose to use 10 hidden nodes without attempting to optimize these parameters. This initial model was adjusted, being the approximation results presented in Figure **4-a** and **4-b**. After the application of the model order reduction methodology, the system was reduced to a third order system for each input ($\overline{\mathbf{A}}$ in Eq. 13 is $[9 \times 9]$). Figure **4-a** and **4-c** shows the reduced model DABNet approximation and the error for the training and test sets, while Figure **4-a** shows the training and test errors for the final reduced DABNet model. It can be appreciated from Figure **4-a** than the approximations of the system output by the initial DABNet model is almost indistinguishable from the final reduced DABNet model. This can be confirmed looking at the approximation errors in Figure **4-b** and **4-c**. It is evident that the nonlinear model performs better than the linear one.

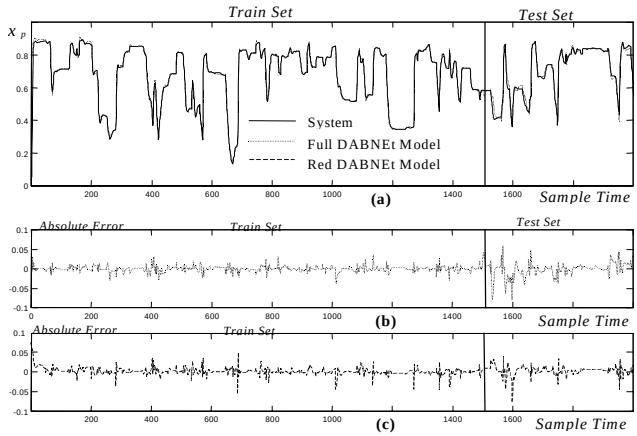


Figure 4: **a)** *DABNet* Model of the Conversion Factor x_p
b) App. Error of the initially full *DABNet* Model $\hat{\mathbf{A}}: [18 \times 18]$
c) App. Error of the reduced *DABNet* Model $\overline{\mathbf{A}}: [9 \times 9]$

5. Conclusions

We have shown a practical approach for identifying nonlinear models from available data and to reducing the dimension of those models. For that purpose we use the DABNet structure, which is composed of a decoupled linear dynamic system followed by nonlinear static map. The linear dynamic system is initially spanned by a set of discrete Laguerre systems, and then cascaded with a single hidden layer Perceptron. A model reduction technique (linear balancing) is performed on the hidden nodes of the neural net. In that way, it is possible to identify the main time constants and to reduce the dimensionality of the Perceptron input. Results concerning the approximation a polymer process were presented, showing the feasibility of this approach in the approximation of typical process units.

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